Estimating the Derivatives

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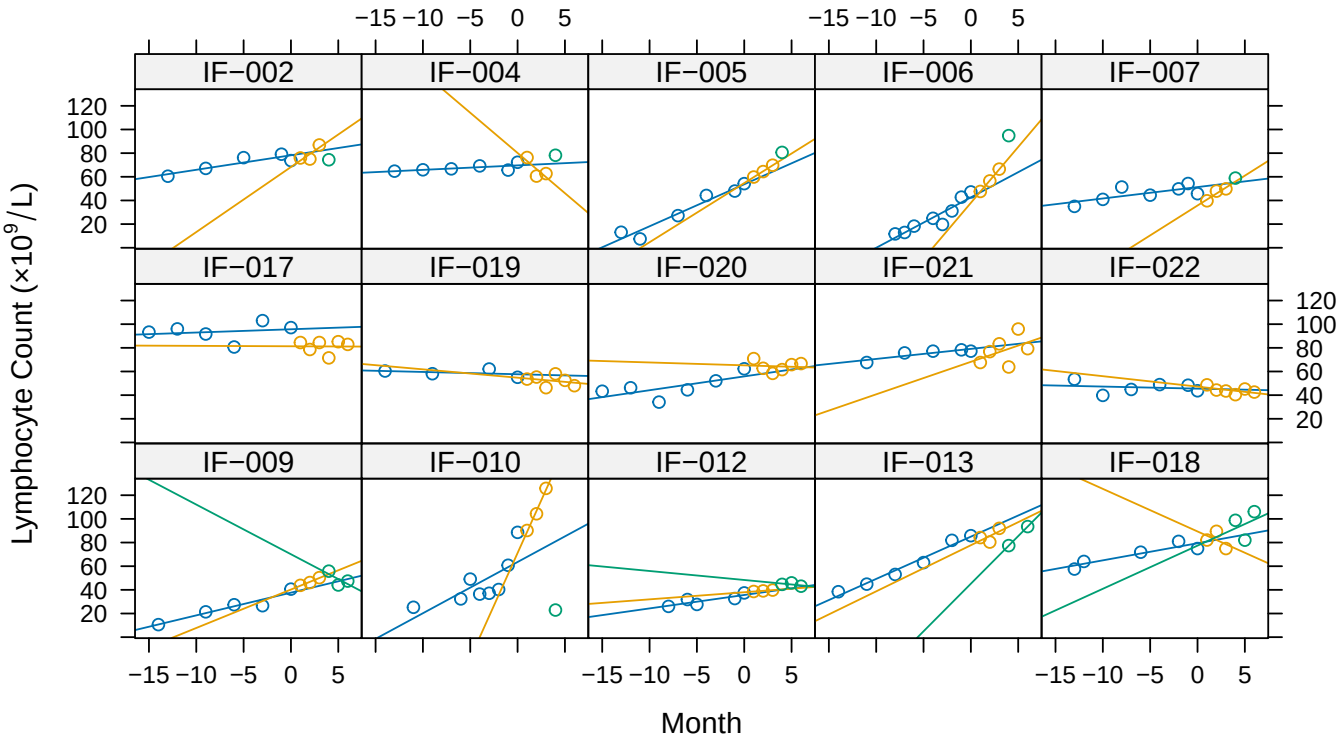
August 1 - 14, 2023 (v1)

1. Data description (same as IF v3.5.Rmd p. 1)

The longitudinal data are from an experiment to study the effect of intermittent fasting (IF) on the lymphocyte measurements of 15 participants, each of whom was repeatedly measured before, during, and after the intervention. The data set (long format) consists of six columns:

- **LYMP** - [numeric] lymphocyte count ($\times 10^9$/L); lower values usually correspond to better outcomes;
- **ATG** - [numeric] relative net LC3-II expression; higher levels are associated with increased autophagy;
- **ID** - [factor] participant identifier; 15 levels;
- **Time** - [numeric] centered time point, with a frequency of 30 days, treating the **last** pretest as Month 0;
- **Time_Within_Phase** - [numeric] time point within each phase, starting from Month 0;
- **Phase** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest.

```
df <- read.csv("~/Downloads/IF_v3.1.csv", stringsAsFactors=T)
lattice::xyplot(LYMP ~ Time | ID, data=df, group=Phase, type=c("p","r"), layout=c(5,3),
                index.cond=list(c(6:9,11,10,12:15,1:5)), #change panel order
                xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
```



2. Modeling

The hypothesis is that IF can help patients lower their lymphocyte count. The objectives are then to fit a curve for each subject's measurement and estimate the first and second derivatives of the function at Month 0 (representing the rate of change and the rate of change of the rate of change).

- First Derivative

Positive derivatives indicate the function is increasing, and negative derivatives indicate the function is decreasing.

- Second Derivative

If the second derivative is positive at a point, the curve is concave upward (e.g., ∪-shape); if the second derivative is negative, the curve is concave downward (e.g., ∩-shape).

A point is a local minimum where the first derivative is zero and the second derivative is positive. Similarly, if the first derivative is zero and the second derivative is negative, that point is a local maximum. If the second derivative is zero and changes sign at a point, that point is an **inflection point** where the curve changes concavity.

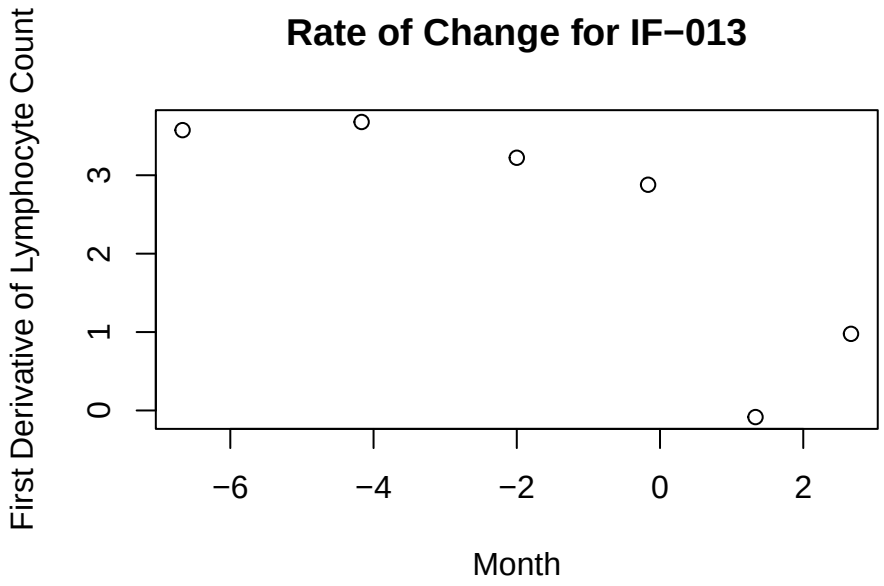
2.1 Using the "*doremi*" package (v 1.0.0)

The *calculate.gold* function calculates the *n*th derivative at some measurement points (called `embeddings`) using the generalized orthogonal local derivative (GOLD) method, which is applicable to irregularly spaced data.

```
dat <- df[!is.na(df$LYMP),] #IF-013 contains one missing value
s13 <- subset(dat, ID=="IF-013")

der_est <- doremi::calculate.gold(time=s13$Time, signal=s13$LYMP, embedding=6, n=2)

plot(der_est$dtime, der_est$dsignal[,2], main="Rate of Change for IF-013",
     xlab="Month", ylab="First Derivative of Lymphocyte Count")
```



2.2 Using the default "*stats*" package (v 4.3.1)

The *smooth.spline* function involves fitting a cubic smoothing spline to the data and then evaluating the derivative of the spline. **Note** that the fitted curve will be different from the GOLD one, but the same as the *calculate.fda* function.

```
spline_fit <- smooth.spline(s13$Time, s13$LYMP) #fit a spline
der_est_1 <- predict(spline_fit, deriv=1)$y #compute the first derivative
# der_est <- doremi::calculate.fda(time=s13$Time, signal=s13$LYMP) #same
names(der_est_1) <- s13$Time
der_est_1
```

```
##      -14      -11       -8       -5       -2        0        1        2
## 3.126697 3.161885 3.214658 3.139196 2.761960 2.390951 2.230657 2.119399
##        3        4        6
## 2.050824 2.030158 2.042499
```

3. Summary

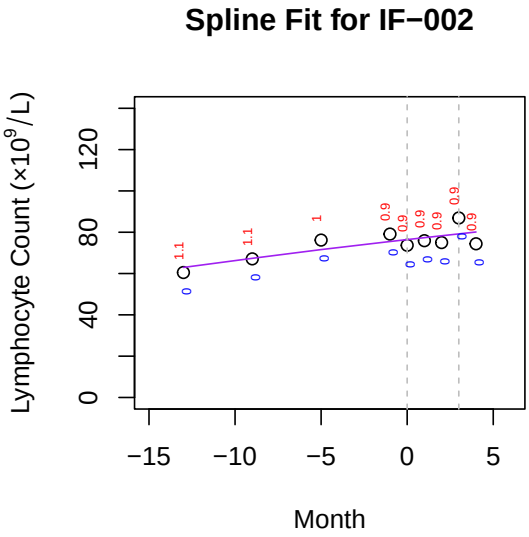
Fit a smoothing spline to each subject's measurement using the *smooth.spline* function.

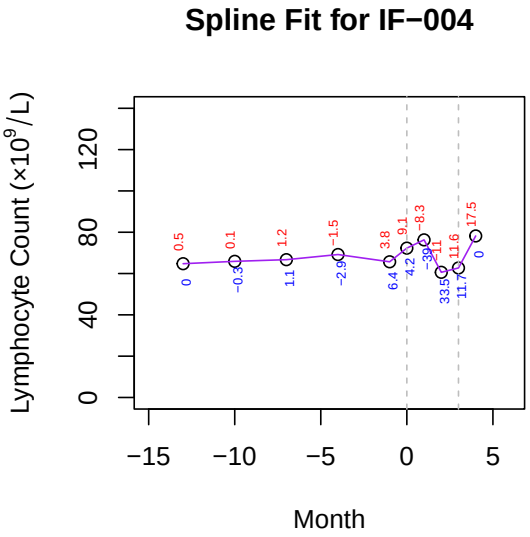
The red text above the point and the blue text below the point indicate the first and second derivatives of the fitted spline at that point, respectively.

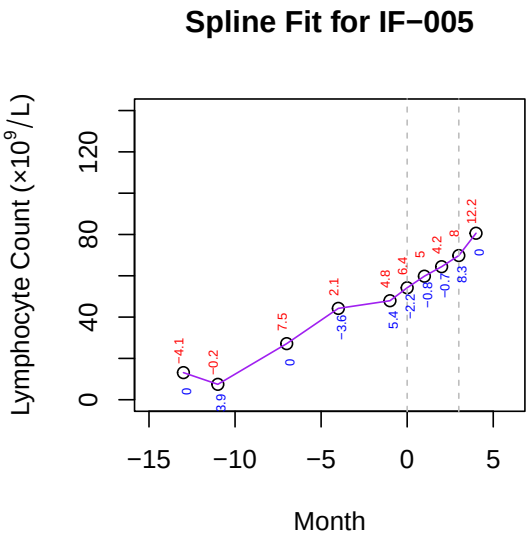
```
par(mfrow=c(1,2))
for (subj in unique(dat$ID)) {
  sub <- subset(dat, ID==subj)
  phases <- c(tail(subset(sub, Phase==0)[,"Time"],1), #start of fasting
              tail(subset(sub, Phase==1)[,"Time"],1)) #start of posttest

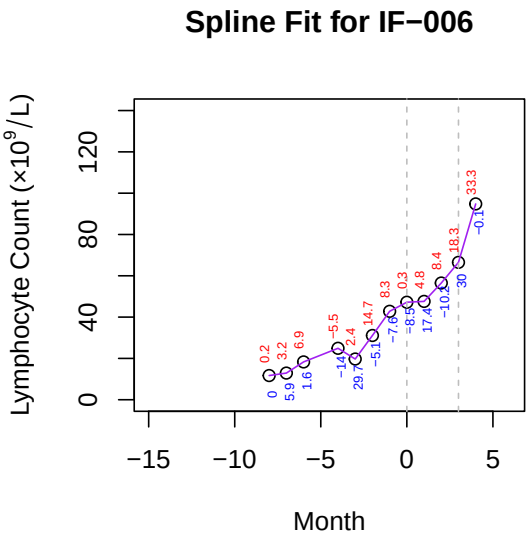
  fit <- smooth.spline(sub$Time, sub$LYMP) #fit a spline
  der1 <- predict(fit, deriv=1)$y #compute the first derivative
  der2 <- predict(fit, deriv=2)$y #compute the second derivative

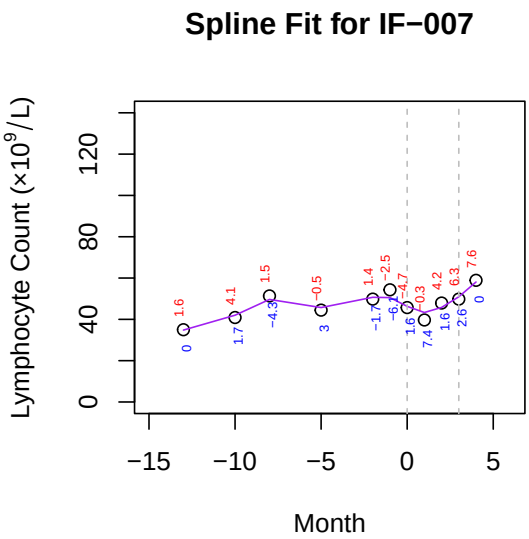
  par(mar=c(5,5,5,5))
  plot(sub$Time, sub$LYMP, ylim=c(0,140), xlim=c(-15,6),
       main=paste0("Spline Fit for ", subj),
       xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
  lines(fit, col="purple")
  abline(v=phases, col="gray", lty=2)
  text(sub$Time, sub$LYMP, labels=round(der1, 1),
       pos=3, cex=.5, offset=.7, srt=90, col="red")
  text(sub$Time, sub$LYMP, labels=round(der2, 1),
       pos=1, cex=.5, offset=.6, srt=90, col="blue")
}
```

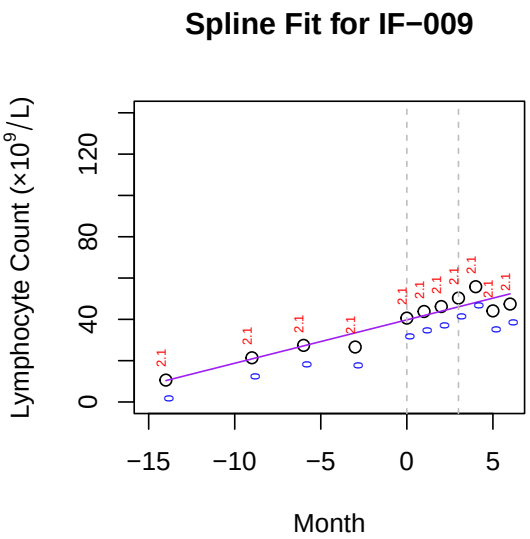

Spline Fit for IF−002
Lymphocyte Count (×10^9/L)
0
40
80
120
−15
−10
−5
0
5
Month

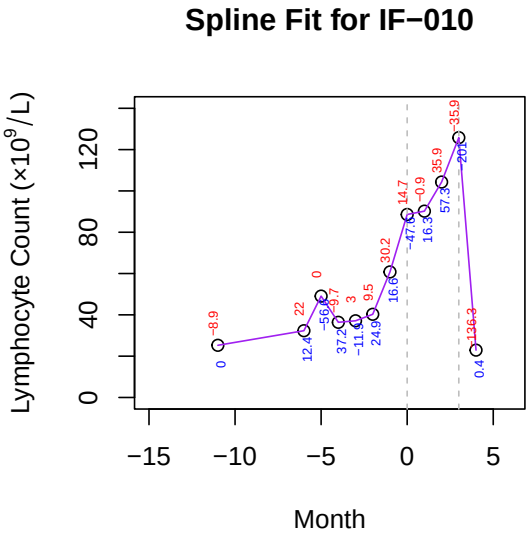

Spline Fit for IF−004
Lymphocyte Count (×10^9/L)
0
40
80
120
−15
−10
−5
0
5
Month

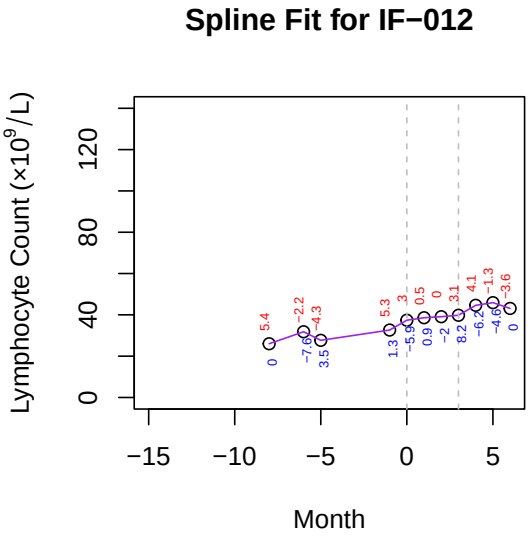

Spline Fit for IF−005
Lymphocyte Count (×10^9/L)
0
40
80
120
−15
−10
−5
0
5
Month

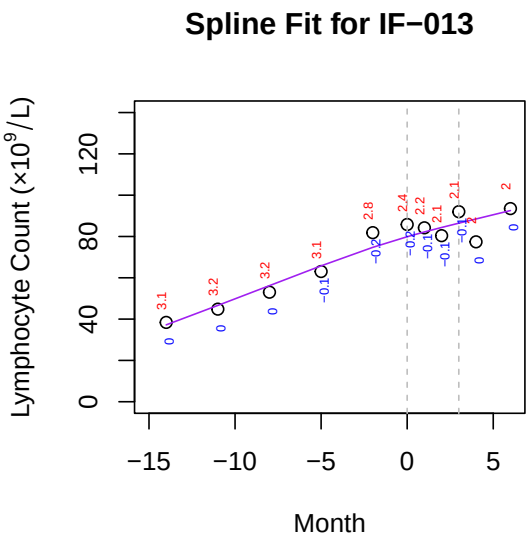

Spline Fit for IF−006
Lymphocyte Count (×10^9/L)
0
40
80
120
−15
−10
−5
0
5
Month

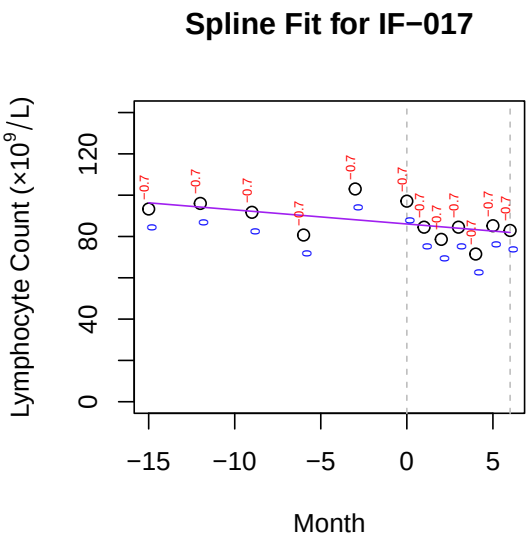

Spline Fit for IF−007
Lymphocyte Count (×10^9/L)
0
40
80
120
−15
−10
−5
0
5
Month

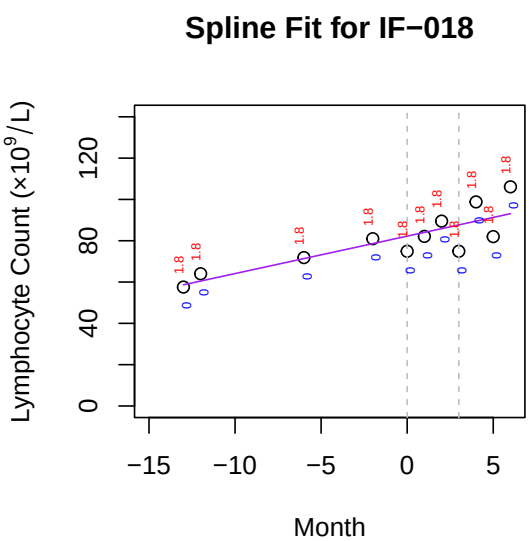

Spline Fit for IF−009
Lymphocyte Count (×10^9/L)
0
40
80
120
−15
−10
−5
0
5
Month

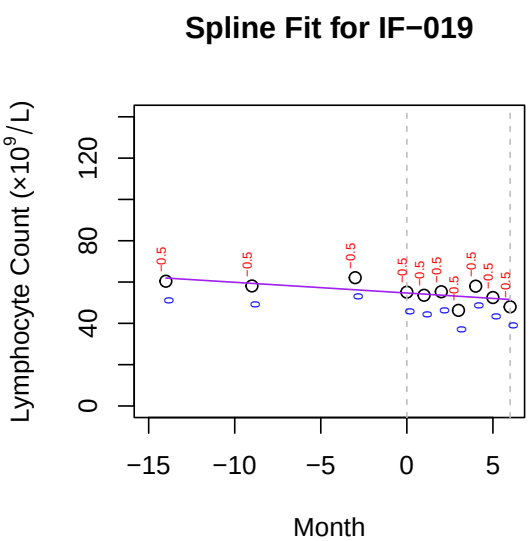

Spline Fit for IF−010
Lymphocyte Count (×10^9/L)
Month

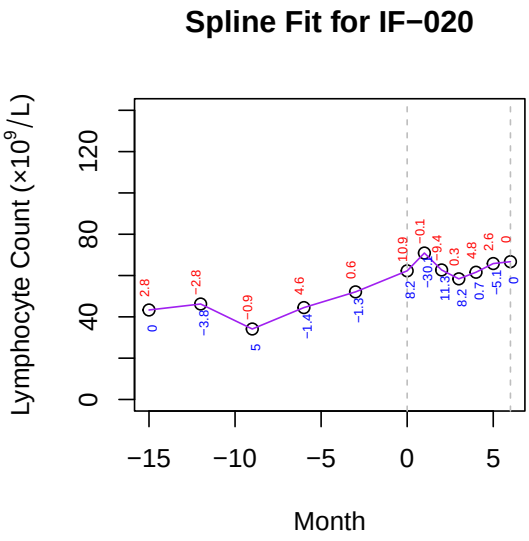

Spline Fit for IF−012
Lymphocyte Count (×10^9/L)
Month

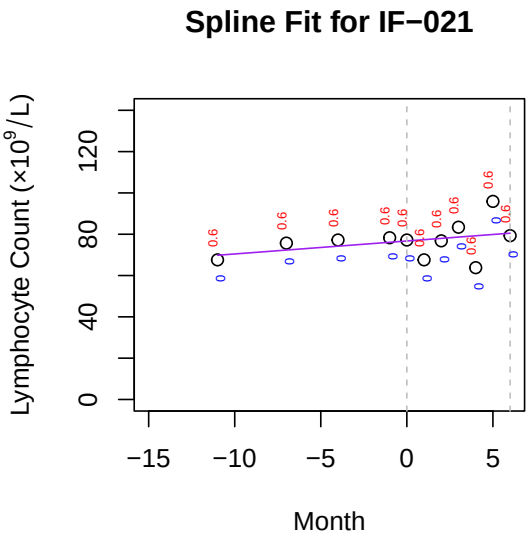

Spline Fit for IF−013
Lymphocyte Count (×10^9/L)
Month

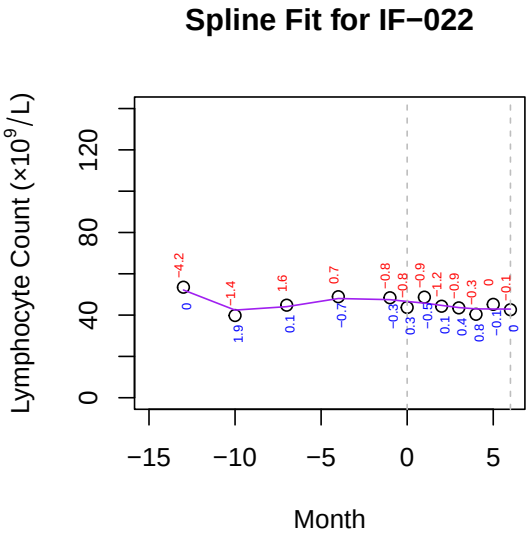

Spline Fit for IF−017
Lymphocyte Count (×10^9/L)
Month


Spline Fit for IF−018
Lymphocyte Count (×10^9/L)
Month


Spline Fit for IF−019
Lymphocyte Count (×10^9/L)
Month







References

Denis, M., Adriana, U., & Delphine, C. (2021). Estimation of the derivatives. https://cran.r-project.org/web/packages/doremi/vignettes/derivatives.html